

TIME SERIES ANALYSIS

Complete Study Guide for MSc Statistics Interviews

24 Comprehensive Chapters

150+ Interview Questions | 100+ Worked Examples

50+ Diagrams & Visualizations | Complete Formula Sheets

From Zero to Interview-Ready

ARIMA

ACF/PACF

Stationarity

Forecasting

White Noise

SARIMA

Designed for: Non-Statistics Undergraduates → MSc Statistics Entrance

Covers all concepts from scratch | Simple Language | Visual First Approach

TABLE OF CONTENTS

1	Introduction to Time Series	What, Why, and Where
2	Components of Time Series	Trend, Seasonality, Cyclical, Irregular
3	Time Series Models	Additive vs Multiplicative
4	Time Series Visualization	Plots, Moving Averages, Smoothing
5	Moving Averages	SMA, WMA, CMA — Solved Examples
6	Exponential Smoothing	SES, Holt's, Holt-Winters
7	Stationarity ■	Most Important — Weak/Strong/Strict
8	Differencing	First, Second, Seasonal Differencing
9	Autocorrelation (ACF) ■	Most Important Interview Topic
10	Partial Autocorrelation (PACF)	PACF vs ACF, Graph Reading
11	White Noise	Definition, Properties, Examples
12	Random Walk	Properties, Drift, Stock Prices
13	AR Models	AR(1), AR(2), AR(p) — Full Derivation
14	MA Models	MA(1), MA(2), MA(q)
15	ARMA Models	Combining AR and MA
16	ARIMA ■■	Most Important — Full p,d,q Guide
17	Seasonal ARIMA (SARIMA)	P,D,Q,s — Seasonal Parameters
18	Model Identification	ACF/PACF Decision Framework
19	Forecasting	One-step, Multi-step, Prediction Intervals
20	Forecast Accuracy	MAE, MSE, RMSE, MAPE
21	Time Series in Statistics	Connections with Regression, Probability
22	Advanced MSc Topics	ADF, KPSS, Cointegration, Unit Root
23	150+ Interview Questions	Conceptual, Numerical, Tricky
24	Complete Cheat Sheet	All Formulas, Definitions, Quick Revision

CHAPTER 1

Introduction to Time Series

Understanding the foundation of temporal data

1.1 What is Time Series?

A **time series** is a sequence of data points collected or recorded at **successive points in time**, usually at **equally spaced intervals**. Think of it like a diary of numbers — every day you write down a number, and together these numbers tell a story that changes over time.

■ Pro Tip

- Simple Definition: Time Series = Data + Time Order
- Key idea: The ORDER of observations MATTERS. Shuffling destroys meaning.
- Each observation depends on what happened BEFORE it — this is what makes it special.

Formal Definition: Let $\{Y_t\} = \{Y_1, Y_2, \dots, Y_T\}$ be a sequence of random variables indexed by time $t = 1, 2, \dots, T$. This sequence is called a **time series**. The subscript t denotes the time period.



1.2 Cross-Sectional vs Time Series Data

To understand time series, let's compare it with the most common type of data you already know — **cross-sectional data**.

Feature	Cross-Sectional Data	Time Series Data
What it is	Data on multiple subjects at ONE point in time	Data on ONE subject at MULTIPLE time points
Order matters?	NO — rows can be shuffled freely	YES — time order is critical
Example	Heights of 100 students in 2024	Daily temperature in Pune, Jan–Dec 2024
Analysis goal	Compare across subjects	Analyse patterns over time

Correlation	Between variables	Between time lags (autocorrelation)
Statistical method	Regression, ANOVA, Clustering	ARIMA, Exponential Smoothing, etc.

1.3 Real-World Examples of Time Series

- **Stock Prices:** NSE NIFTY 50 closing price — recorded every trading day. Pattern: random walk with upward drift over long term.
- ■ **Temperature:** Daily maximum temperature in Pune — shows seasonal pattern (hot summers, cooler winters).
- ■ **Rainfall:** Monthly rainfall in Maharashtra — strong seasonal pattern, low in winter, high in June-September.
- **Population:** India's annual population — shows steady upward trend with no seasonality.
- **GDP:** Quarterly GDP growth — shows trend, business cycles, and seasonal effects.
- **Sales Data:** Monthly sales of umbrellas — obvious seasonal spike every June-August.
- **Heart Rate:** Patient's heart rate every minute — used in healthcare monitoring.
- **Electricity Demand:** Daily power consumption — peaks in mornings and evenings, seasonal patterns.

1.4 Why is Time Series Important?

Time series analysis is one of the most practically important branches of statistics. Here is **why every statistician and data scientist must know it**:

Field	Application	What Time Series Enables
Finance	Stock market prediction, Risk management	Forecast future prices, detect volatility patterns
Economics	GDP forecasting, Inflation analysis	Policy decisions based on economic trends
Healthcare	Epidemic modelling, Patient monitoring	Detect anomalies, predict disease spread
Meteorology	Weather forecasting	Predict rain, temperature, disasters
Supply Chain	Demand forecasting, Inventory management	Reduce overstock and stockouts
Data Science	ML time series models, Feature engineering	Build predictive systems with temporal data
Engineering	Signal processing, Quality control	Detect equipment failures in advance

1.5 Key Vocabulary You Must Know

Term	Meaning	Simple Example
Time Series	Data ordered by time	Daily temperatures
Observation (Y_t)	Data value at time t	Temperature on Day 5 = 32°C
Lag	A previous time point	Yesterday's value = Lag 1

Forecast	Predicted future value	Tomorrow's temperature = 33°C
Residual	Actual minus predicted	32 - 33 = -1 (forecast error)
Stationarity	Statistical properties don't change over time	Constant mean, variance
Trend	Long-term direction (up or down)	Rising GDP over 10 years
Seasonality	Regular pattern that repeats at fixed intervals	Ice cream sales peak every summer

■ Interview Questions

- Q: What is a time series? Give an example. [30-second answer: A sequence of data points collected over time at regular intervals.]
- Q: How is time series data different from cross-sectional data? [Key answer: In time series, ORDER matters. You cannot swap observations.]
- Q: Why can't we use ordinary regression for time series data? [Because observations are NOT independent — each value is influenced by the previous one.]
- Q: Give 3 real-world applications of time series analysis. [Finance: stock prediction; Weather: temperature forecasting; Retail: sales forecasting.]
- Q: What is the difference between a time series and a random variable? [A random variable is a single measurement. A time series is a sequence of measurements over time.]

■ Memory Tricks

- Think of time series as a STORY — the plot matters, and you can't read the chapters out of order!
- T.S. = Time + Sequence. Remember: TIME SEQUENCE.
- Cross-sectional = snapshot (one moment, many people). Time series = movie (one thing, many moments).

CHAPTER 2

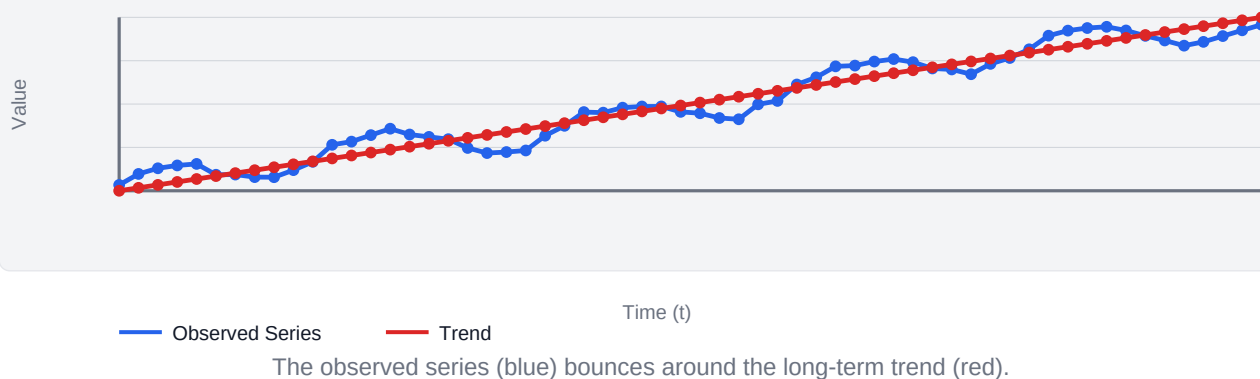
Components of Time Series

Decomposing a time series into its building blocks

2.1 The Four Components — Overview

Every time series can be broken down into **four underlying components**. Understanding these is the **FIRST** step in any time series analysis. Think of it like taking apart a cake to understand its ingredients.

Figure 2.1 — Observed Series with Underlying Trend



2.2 Trend (T)

Trend is the **long-term movement** or direction in the data. It shows whether the data is generally going **up** (upward trend), **down** (downward trend), or **staying flat** (no trend) over a long period.

Key characteristics of Trend:

- It is **SMOOTH** — short-term fluctuations are ignored.
- It operates over a **LONG** time horizon (years or decades).
- It can be linear (straight line) or non-linear (curved).
- Caused by factors like population growth, economic development, technological change.

Examples of Trends:

Type	Example	Cause
Upward Linear Trend	India's GDP growing steadily	Economic development
Downward Trend	Landline phone usage declining	Replaced by mobile phones
Non-linear Trend	Internet adoption (S-curve growth)	Technology diffusion
No Trend (flat)	Temperature around a constant mean	Stationary data

2.3 Seasonality (S)

Seasonality refers to **regular, predictable fluctuations** that repeat at **fixed intervals** — usually within one calendar year (monthly, quarterly). It is caused by external factors like **weather, holidays, or customs**.



Real-world seasonal examples:

- Umbrella sales: spike every June–August (monsoon season in India)
- Retail sales: spike every October–December (festive season, Diwali, Christmas)
- Ice cream sales: peak in summer months (April–June)
- Agricultural output: depends on sowing and harvest seasons
- Tourism: peak in December (winter holidays in Europe)

2.4 Cyclical Variation (C)

Cyclical variation refers to **fluctuations that occur over longer periods** (usually 2–10 years) and do **NOT** have a **fixed, regular period**. These are caused by **business cycles** — boom, recession, recovery.

Feature	Seasonality	Cyclical Variation
Period	Fixed (e.g., every 12 months)	Variable (2–10+ years)
Regularity	Very regular and predictable	Irregular — period changes each time
Cause	Weather, festivals, customs	Business cycles, economic conditions
Duration	Short (within a year)	Long (multiple years)
Predictability	Highly predictable	Harder to predict exactly
Example	Christmas sales spike every December	Stock market boom and bust cycles

2.5 Irregular/Random Variation (I)

Irregular variation (also called **random** or **residual** variation) is the **unpredictable, erratic fluctuation** remaining after removing trend, seasonality, and cyclical components. It represents **pure randomness** and cannot be explained by any systematic pattern.

- Caused by unexpected events: natural disasters, strikes, pandemics, accidents
- Cannot be predicted or forecasted

- This is what remains after all other components are accounted for
- In a good model, the residuals should look like 'white noise' (pure random)

Figure 2.3 — Pure Irregular/Random Component



No pattern here — this is pure randomness (white noise).

2.6 Summary of All Four Components

Component	Symbol	Nature	Duration	Predictable?	Example
Trend	T	Long-term direction	Years/Decades	Yes	Rising population
Seasonal	S	Regular repeating pattern	< 1 year	Yes	Monsoon sales
Cyclical	C	Wave-like oscillation	2–10 years	Partially	Recession cycles
Irregular	I	Random unpredictable noise	Very short	No	Earthquake impact

■ Interview Questions

- Q: What are the four components of a time series? [Trend (T), Seasonality (S), Cyclical (C), Irregular (I). Always remember the acronym TSCIR]
- Q: What is the difference between seasonal and cyclical variation? [Seasonal has FIXED period (within a year, predictable), cyclical has NO fixed period]
- Q: Can a time series have both trend and seasonality? [YES! Most real series have multiple components. E.g., airline passenger numbers]
- Q: What does 'decomposition' mean? [It means separating a time series into its T, S, C, I components to study each separately]
- Q: What is irregular variation? Can it be forecasted? [It's the random, unpredictable residual after removing T, S, C. I]

■ Tricky / Trap Questions

- TRAP: 'Seasonal variation always means summer/winter patterns.' WRONG — seasonality just means any REGULAR pattern
- TRAP: 'Cyclical = Seasonal.' WRONG — cyclical has NO fixed period. Seasonal period is fixed. This distinction is critical
- TRAP: 'Irregular variation is just error.' Partially right — it includes measurement error BUT also genuine random shock

■ Memory Tricks

- TSCI = 'The Story Continues Infinitely' — T(rend) S(easonal) C(yclical) I(rregular)
- Seasonal = CALENDAR (fixed timing each year). Cyclical = ECONOMIC (tied to business cycles, variable length).
- Irregular = 'What's left after explaining everything else.' It's the mystery component.

CHAPTER 3

Time Series Models

Additive vs Multiplicative decomposition models

3.1 Why Do We Need Models?

Once we know that a time series has components (T, S, C, I), we need a mathematical way to **combine** these components to represent the observed data. There are **two main ways** to do this:

$$\text{Additive: } Y_t = T_t + S_t + C_t + I_t$$

$$\text{Multiplicative: } Y_t = T_t \times S_t \times C_t \times I_t$$

3.2 Additive Model

In the **additive model**, the four components are simply **added together**. This means each component acts **independently** of the others. The seasonal effect is **constant in absolute terms** — it adds the same amount regardless of the level of the series.

When to use Additive Model:

- The seasonal fluctuations are **ROUGHLY CONSTANT** over time (e.g., sales always increase by exactly 500 units in December).
- The series does **NOT** grow exponentially — changes are **ABSOLUTE** (not percentage-based).
- Seasonal variation is **INDEPENDENT** of the trend level.

Worked Example — Additive Model:

Suppose quarterly sales (units) for Year 1 with:

Trend $T = 200$ (constant), Seasonal effects $S = [+30, -10, +20, -40]$ for Q1–Q4, No cyclical ($C=0$), Irregular $I = [5, -3, 2, -4]$

Quarter	Trend (T)	Seasonal (S)	Irregular (I)	Observed $Y = T+S+I$
Q1	200	+30	+5	$200+30+5 = 235$
Q2	200	-10	-3	$200-10-3 = 187$
Q3	200	+20	+2	$200+20+2 = 222$
Q4	200	-40	-4	$200-40-4 = 156$

3.3 Multiplicative Model

In the **multiplicative model**, the components are **multiplied together**. The seasonal effect is **proportional to the trend level** — so as the series grows, the seasonal swings also grow LARGER. This is more realistic for economic and business data.

When to use Multiplicative Model:

- Seasonal fluctuations INCREASE as the trend increases (e.g., sales increase by 20% every December, not a fixed amount).
- The series grows exponentially or proportionally over time.
- The amplitude of seasonal variation is NOT constant — it grows or shrinks with the series level.

Worked Example — Multiplicative Model:

Quarterly sales with Trend $T = [200, 210, 220, 230]$ and Seasonal indices $S = [1.15, 0.95, 1.10, 0.80]$:

Quarter	Trend (T)	Seasonal Index (S)	Calculated $Y = T \times S$
Q1 (Yr1)	200	1.15	$200 \times 1.15 = 230$
Q2 (Yr1)	210	0.95	$210 \times 0.95 = 199.5$
Q3 (Yr1)	220	1.10	$220 \times 1.10 = 242$
Q4 (Yr1)	230	0.80	$230 \times 0.80 = 184$

Notice: In Q4, the seasonal index is 0.80, meaning sales are 20% BELOW trend. As the trend grows, the absolute drop in Q4 also grows — this is the multiplicative effect.

3.4 Additive vs Multiplicative — Decision Guide

Aspect	Additive Model	Multiplicative Model
Formula	$Y = T + S + C + I$	$Y = T \times S \times C \times I$
Seasonal effect	Constant (fixed units)	Proportional (% of trend)
Look at the graph	Seasonal bands are EQUAL in width	Seasonal bands WIDEN as trend grows
When trend rises...	Seasonal variation stays the same	Seasonal variation grows proportionally
Series type	Linear growth or no growth	Exponential or rapid growth
Real example	Temperature data	Company revenue data
Transformation	No transformation needed	Take log → becomes additive
Seasonal indices	Sum to zero (or near zero)	Multiply to 1 (e.g., average = 1.0)

■ Pro Tip

- KEY TRICK: Plot the data first! If the seasonal 'bumps' get BIGGER as the series grows, use Multiplicative. If bumps stay the same, use Additive.
- Mathematical link: Taking LOG of a Multiplicative model converts it to Additive: $\log(Y) = \log(T) + \log(S) + \log(C) + \log(I)$

■ Interview Questions

- Q: What is the difference between additive and multiplicative time series models? [In additive, $Y=T+S+C+I$ — components
- Q: How do you decide which model to use? [Look at the graph: if seasonal variation is constant → additive; if it grows w
- Q: How can you convert a multiplicative model to an additive model? [Take the NATURAL LOG: $\log(Y) = \log(T) + \log(S) +$
- Q: In the multiplicative model, what does a seasonal index of 1.20 mean? [It means the value for that period is 20% ABOVE
- Q: What is the sum of seasonal indices in an additive model? [They sum to ZERO (over one full seasonal cycle). In multip

■ Common Mistakes

- Saying 'Additive is always better' — WRONG. Choice depends on the data pattern.
- Confusing seasonal INDEX (multiplicative) with seasonal EFFECT (additive). Index is a ratio (e.g., 1.2), effect is an ab
- Forgetting that log transformation converts multiplicative to additive — very common exam question!

CHAPTER 4

Time Series Visualization

Seeing the patterns before calculating anything

4.1 Why Visualize First?

In time series analysis, **plotting your data is the FIRST step** — always, without exception. A good plot tells you: Is there a trend? Is there seasonality? Are there outliers? Is the variance constant? This visual information guides EVERY decision you make later.

4.2 Line Plot (Time Plot)

The **line plot** is the most fundamental time series plot. It plots Y_t (value) on the y-axis against t (time) on the x-axis and connects points with a line.



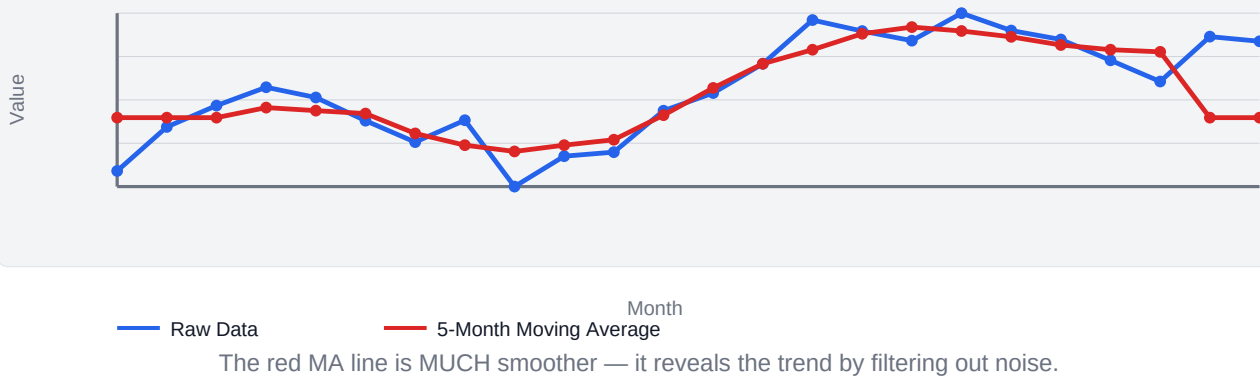
What to look for in a time plot:

What you see	What it means	What to do
General upward/downward direction	TREND present	Use differencing or detrending
Regular repeating peaks/troughs	SEASONALITY present	Apply seasonal adjustment
Spread (variance) increasing	Heteroscedasticity	Log transformation
Sudden jumps or drops	OUTLIERS / structural breaks	Investigate separately
No pattern — random scatter	Possibly stationary / white noise	Check ACF

4.3 Moving Average Smoothing

A **moving average** is a technique to **smooth out short-term fluctuations** and highlight the underlying **trend**. We calculate an average over a 'window' of consecutive observations, then slide that window forward one step at a time.

Figure 4.2 — Raw Data vs 5-Month Moving Average



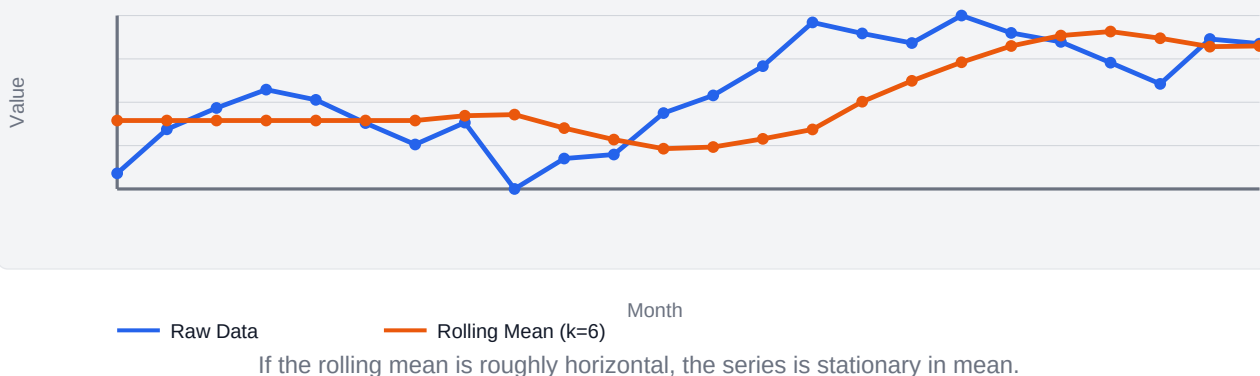
4.4 Rolling Mean and Rolling Variance

Rolling statistics are computed in a moving window and are crucial for checking stationarity:

Rolling Mean: The average of the last k observations. If rolling mean is **approximately constant**, it suggests NO trend (good for stationarity). If it changes over time, trend exists.

Rolling Variance (or Std Dev): The variance of the last k observations. If rolling variance is **approximately constant**, it suggests homoscedasticity (constant spread). If it changes, the series may need a transformation.

Figure 4.3 — Rolling Mean (constant = stationary signal)



4.5 Graph Interpretation for Interviews

Interviewers LOVE to show you a graph and ask 'What do you see?' Here is a systematic framework to answer any graph question:

Step	What to check	Comment out loud
1	Overall direction (trend)	'I can see an upward/downward/no trend...'
2	Regular patterns (seasonality)	'There appears to be a seasonal pattern repeating every...'
3	Spread over time (variance)	'The variance seems constant/increasing/decreasing...'
4	Sudden changes (structural breaks)	'There is a sudden jump around time $t=...$ '

5	Whether it looks stationary	'The series does/does not appear stationary because...'
---	-----------------------------	---

■ Interview Questions

- Q: What is a time plot and what do you look for? [It's a line graph of Y_t vs t . Look for: trend direction, seasonality,
- Q: What does a moving average do to a time series? [It SMOOTHS it — removes short-term noise and highlights the underlying trend]
- Q: If the rolling mean is increasing in a plot, what does it tell you? [The series has a TREND and is likely NON-STATIONARY]
- Q: What is the disadvantage of a moving average? [It loses data points at the ends (no MA for first and last window/2 points)]

CHAPTER 5

Moving Averages

SMA, WMA, CMA — Formulas, Solved Problems, Forecasting

5.1 Simple Moving Average (SMA)

The **Simple Moving Average** of order k at time t is the **arithmetic mean** of the k most recent observations. It gives **equal weight** to all k observations.

$$\text{SMA}(k)_t = (Y_{\{t\}} + Y_{\{t-1\}} + \dots + Y_{\{t-k+1\}}) / k$$

■ Worked Example — SMA:

Monthly sales (units): 100, 120, 110, 130, 140, 125, 135. Calculate 3-month SMA.

Month (t)	Sales (Y _t)	3-Month SMA = (Y _t + Y _{t-1} + Y _{t-2})/3	Calculation
1	100	—	Not enough data
2	120	—	Not enough data
3	110	110.0	(100+120+110)/3 = 110.0
4	130	120.0	(120+110+130)/3 = 120.0
5	140	126.7	(110+130+140)/3 = 126.7
6	125	131.7	(130+140+125)/3 = 131.7
7	135	133.3	(140+125+135)/3 = 133.3

Interpretation: The SMA at month 3 is 110.0 — this means the average sales over the last 3 months is 110 units. We FORECAST month 8 as the last SMA = 133.3 units.

5.2 Weighted Moving Average (WMA)

The **Weighted Moving Average** assigns **different weights** to observations, with more recent observations typically getting **higher weights**. This is more flexible than SMA because recent data can be made more influential.

$$\text{WMA}_t = (w_1 \cdot Y_t + w_2 \cdot Y_{\{t-1\}} + \dots + w_k \cdot Y_{\{t-k+1\}}) / (w_1 + w_2 + \dots + w_k)$$

where $w_1 \geq w_2 \geq \dots \geq w_k$ (recent observations get higher weights)

■ Worked Example — WMA:

Monthly sales: 100, 120, 110, 130. Weights: 3 (most recent), 2, 1 (oldest).

Calculate WMA for month 4: $WMA = (3 \times 130 + 2 \times 110 + 1 \times 120) / (3+2+1) = (390+220+120)/6 = 730/6 = \mathbf{121.7}$

Compare: $SMA(3)$ for month 4 = $(120+110+130)/3 = 120.0$

The WMA (121.7) gives more importance to the most recent value of 130, pulling the estimate higher.

5.3 Centered Moving Average (CMA)

The **Centered Moving Average** is used when the order k is **even** (e.g., $k=4$ or $k=12$). For odd k , the MA naturally centers at the middle observation. For even k , we take a second moving average of the moving averages to center it.

CMA: Average of two consecutive 4-point MAs → centered at integer time point

■ Worked Example — CMA (order 4):

Quarterly data: Q1=40, Q2=55, Q3=65, Q4=48, Q1=42, Q2=58, Q3=70, Q4=50

Quarter (t)	Value	4-pt MA	Centered MA (CMA)
Q1 (t=1)	40	—	—
Q2 (t=2)	55	—	—
Q3 (t=3)	65	$(40+55+65+48)/4 = 52.0$	$(52.0+52.5)/2 = 52.25$
Q4 (t=4)	48	$(55+65+48+42)/4 = 52.5$	$(52.5+53.75)/2 = 53.125$
Q1 (t=5)	42	$(65+48+42+58)/4 = 53.25$	$(53.25+55.5)/2 = 54.375$
Q2 (t=6)	58	$(48+42+58+70)/4 = 54.5$	$(54.5+55.0)/2 = 54.75$
Q3 (t=7)	70	$(42+58+70+50)/4 = 55.0$	—
Q4 (t=8)	50	—	—

■ Pro Tip

- CMA is used in the DECOMPOSITION method to extract the trend-cycle component from quarterly or monthly data.
- Why center? Because a 4-point MA would fall BETWEEN time points (e.g., between Q2 and Q3). Centering aligns it with a

5.4 Advantages and Disadvantages

Type	Advantages	Disadvantages
SMA	Simple to calculate; good for stable series	Lags behind actual data; equal weight is unrealistic; loses end data
WMA	More flexible; recent data gets more weight	Choosing weights is subjective; more complex
CMA	Best for seasonal decomposition; properly centered	Cannot compute at the extremes; requires 2 passes

5.5 Using MA for Forecasting

For forecasting, the **SMA is a naive but useful method**: The forecast for the next period is simply the current SMA.

$$\text{Forecast: } \hat{Y}_{t+1} = \text{SMA}(k)_t = (Y_t + Y_{t-1} + \dots + Y_{t-k+1}) / k$$

■ Interview Questions

- Q: What is a moving average? [An average calculated over a sliding window of k observations, used to smooth data and remove noise.]
- Q: What is the difference between SMA and WMA? [SMA gives equal weights to all k observations. WMA assigns higher weights to more recent observations.]
- Q: Why do we use a Centered Moving Average? [When order k is even, the simple MA falls between time points. Centering the MA aligns it with the time points.]
- Q: What is the main weakness of moving averages for forecasting? [They LAG behind the actual data. In a trending series, the forecast will be lower than the actual values.]
- Q: How many data points are 'lost' in a 5-period moving average? [2 at the beginning and 2 at the end = 4 total lost. Generally, $k-1$ points are lost.]

■ Common Mistakes

- Forgetting to center the MA when order is EVEN. Always use CMA for even-order MAs in decomposition.
- Using MA to forecast a seasonal series without adjusting for seasonality — MA smooths out seasonality, so the forecast will be lower than the actual values.
- Choosing too large a window: over-smoothing hides useful patterns. Too small: doesn't remove enough noise.

CHAPTER 6

Exponential Smoothing

SES, Double ES, Holt's, and Holt-Winters Methods

6.1 Why Exponential Smoothing?

Moving averages have a major flaw: they give **equal weight** to old and recent data. In reality, **recent observations are more relevant**. Exponential Smoothing fixes this by giving **exponentially decreasing weights** to past observations — recent data gets the most weight, older data gets less and less.

6.2 Simple Exponential Smoothing (SES)

SES is used for series with **no trend** and **no seasonality**. It produces a smoothed value S_t which is also the one-step-ahead forecast.

$$S_t = \alpha \cdot Y_t + (1-\alpha) \cdot S_{t-1}$$

where α (**alpha**) is the **smoothing parameter**, $0 < \alpha < 1$.

Understanding α (Alpha):

Value of α	Effect	When to use
α close to 1 (e.g., 0.9)	High weight on recent data, low memory	Rapidly changing series, recent data is most informative
α close to 0 (e.g., 0.1)	Low weight on recent data, high memory	Stable series, want smooth output, past data still relevant
$\alpha = 0.5$	Equal balance between old and new	Moderately changing series

■ Worked Example — SES with $\alpha = 0.3$:

Sales data: 50, 55, 58, 62, 60. Initial $S_1 = Y_1 = 50$.

t	Y_t	$S_t = 0.3 \cdot Y_t + 0.7 \cdot S_{t-1}$	Forecast $\hat{Y}_{t+1} = S_t$
1	50	50.0 (initialise)	50.0
2	55	$0.3 \times 55 + 0.7 \times 50 = 16.5 + 35 = 51.5$	51.5
3	58	$0.3 \times 58 + 0.7 \times 51.5 = 17.4 + 36.05 = 53.45$	53.45
4	62	$0.3 \times 62 + 0.7 \times 53.45 = 18.6 + 37.42 = 56.02$	56.02
5	60	$0.3 \times 60 + 0.7 \times 56.02 = 18 + 39.21 = 57.21$	57.21
6 (Forecast)	—	—	Forecast = 57.21

6.3 Double Exponential Smoothing / Holt's Method

SES cannot handle a **trend**. **Holt's Method** (Double Exponential Smoothing) adds a second equation to track the trend. It uses **two parameters**: α and β .

$$\text{Level: } L_t = \alpha \cdot Y_t + (1-\alpha) \cdot (L_{t-1} + T_{t-1})$$

$$\text{Trend: } T_t = \beta \cdot (L_t - L_{t-1}) + (1-\beta) \cdot T_{t-1}$$

$$\text{Forecast } h \text{ steps ahead: } \hat{Y}_{t+h} = L_t + h \cdot T_t$$

where β (**beta**) is the trend smoothing parameter ($0 < \beta < 1$). L_t is the estimated level, T_t is the estimated trend.

■ Worked Example — Holt's Method ($\alpha=0.4$, $\beta=0.3$):

Sales: 200, 210, 225, 240. Initialize: $L_1=200$, $T_1=10$.

t	Y_t	L_t	T_t	1-step Forecast
1	200	200.0 (init)	10.0 (init)	—
2	210	$0.4 \times 210 + 0.6 \times (200 + 10) = 84 + 126 = 210$	$0.3 \times (210 - 200) + 0.7 \times 10 = 3 + 7 = 10$	$200 + 1 \times 10 = 210.0$
3	225	$0.4 \times 225 + 0.6 \times (210 + 10) = 90 + 132 = 222$	$0.3 \times (222 - 210) + 0.7 \times 10 = 3.6 + 7 = 10.6$	$210 + 1 \times 10 = 220.0$
4	240	$0.4 \times 240 + 0.6 \times (222 + 10.6) = 96 + 139.6 = 235.6$	$0.3 \times (235.6 - 222) + 0.7 \times 10.6 = 4.08 + 7.42 = 11.5$	$222 + 1 \times 10.6 = 232.6$
5 (Forecast)	—	—	—	$235.6 + 1 \times 11.5 = 247.1$

6.4 Holt-Winters Method (Triple Exponential Smoothing)

For series with **both trend AND seasonality**, we use the **Holt-Winters method**, which adds a third equation for seasonality. Uses **three parameters**: α , β , γ .

Parameter	Symbol	Controls	Effect when close to 1	Effect when close to 0
Alpha	α	Level smoothing	High responsiveness to recent Y_t	More weight on past level
Beta	β	Trend smoothing	Trend adapts quickly to new data	Trend is very stable/slow to change
Gamma	γ	Seasonal smoothing	Seasonal indices update quickly	Seasonal indices are very stable

$$\text{Additive HW: } \hat{Y}_{t+h} = (L_t + h \cdot T_t) + S_{t+h-m}$$

$$\text{Multiplicative HW: } \hat{Y}_{t+h} = (L_t + h \cdot T_t) \times S_{t+h-m}$$

where m = length of seasonal cycle (e.g., $m=12$ for monthly, $m=4$ for quarterly)

6.5 Choosing Between SES, Holt's, and Holt-Winters

Method	Trend?	Seasonality?	Parameters	Use Case
Simple ES (SES)	No	No	α only	Stable series around a constant level
Holt's Method	Yes	No	α, β	Series with trend but no seasonality
Holt-Winters (Additive)	Yes	Yes (constant amplitude)	α, β, γ	Series with both, constant seasonal swings
Holt-Winters (Multiplicative)	Yes	Yes (growing amplitude)	α, β, γ	Series with growing seasonal swings

■ Interview Questions

- Q: What is exponential smoothing? [A forecasting method that gives exponentially decreasing weights to past observations]
- Q: What does the parameter α control in SES? [It controls the balance between the current observation and past smoothed values]
- Q: What is the difference between SES and Holt's Method? [SES handles only stationary series (no trend). Holt's adds a trend component]
- Q: What does γ (gamma) control in Holt-Winters? [It controls how quickly the seasonal indices are updated. High γ = seasonal changes are updated quickly]
- Q: If $\alpha = 1$ in SES, what happens? [The forecast for the next period is just the LAST observed value (naive forecast). $S_t = y_t$]

■ Memory Tricks

- SES = No trend, no season (ONE component smoothed, ONE parameter α)
- Holt's = Trend only (TWO components smoothed, TWO parameters α, β)
- Holt-Winters = Trend + Season (THREE components, THREE parameters α, β, γ)
- Acronym: 'Some Hot Women' = SES, Holt, Holt-Winters ■